

## Multiple Choice (10 marks)

1. Which of the following was not cited as evidence for life in the martian meteorite ALH 84001?
  - (a) Complex organic molecules specifically PAHs
  - (b) Magnetite grains similar to those formed by bacteria on Earth
  - (c) Carbonate minerals indicating a thicker warmer Martian atmosphere
  - (d) Amino acids with a preferred orientation or "chirality"
2. What Mars mission will be landing on May 25 2008 and will dig a trench into (hopefully) ice-rich soil?
  - (a) ExoMars
  - (b) Mars Exploration Rovers
  - (c) Mars Science Laboratory
  - (d) Phoenix Mars Lander
3. You observe the visible spectrum of a star viewed through a cloud of cool hydrogen gas.
  - (a) You see only the star's blackbody spectrum.
  - (b) You see the star's blackbody spectrum with absorption lines due to hydrogen.
  - (c) You see only emission lines characteristic of hydrogen.
  - (d) You see only emission lines characteristic of the star's composition.
4. Which of the following countries has sent landers to Venus?
  - (a) The U.S.S.R.
  - (b) The U.S.
  - (c) France
  - (d) A and B
5. By locating the north celestial pole (NCP) in the sky how can you determine your latitude?
  - (a) The azimuth of the NCP is the angular distance from the North Pole.
  - (b) The azimuth of the NCP is the same as your latitude.
  - (c) The altitude of the NCP is the same as your latitude.
  - (d) The altitude of the NCP is your angular distance from the North Pole.

## True / False (10 marks)

*Answer True or False*

6. True or False: The Large Magellanic Cloud is the closest planetary nebula to Earth.
7. True or False: Type Ia supernovae occur in binary systems, where a white dwarf accretes material from a companion star until it reaches a critical mass.
8. True or False: The length of a year would remain unchanged because the gravitational force from the Sun would still be the same.
9. True or False: Sunspots appear in cycles approximately every 9 years, which is referred to as the solar cycle.
10. True or False: The order of electromagnetic radiation from shortest to longest wavelength is infrared, visible light, ultraviolet, X rays, gamma rays, and radio.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the role of gravitational resonance with Jupiter in preventing the formation of a planet in the asteroid belt, and analyze the implications for our understanding of planetary formation.
12. Discuss the reasons for the current tilt of the Mars Exploration Rover Spirit towards the north, considering its location in the southern hemisphere during winter.

*End of Paper*